Disentangling Debiasing from Abstention in Bias-Steering Transfer to Quantized Language Models

Aldrin Kabya Biswas, Md. Zakir Hossan, Md. Ashraful Amin, and AKM Mahbubur Rahman

Center for Computational & Data Sciences, Independent University, Bangladesh, Dhaka-1229, Bangladesh
aldrin.kabya04@gmail.com, {zakir,aminmdashraful,akmmrahman}@iub.edu.bd

Abstract. Activation steering reduces social bias in large language models by adding a fixed direction, a steering vector, to the model's hidden states at inference time. Such vectors are usually extracted from the unquantized (FP16) model, whereas most deployed models are quantized. We ask whether a vector extracted once on FP16 still works when added unchanged to quantized versions of the model, and whether it reduces bias or merely makes the model decline to answer. On the BBQ benchmark, a model can lower its bias score simply by answering "unknown" more often, since that is the correct answer to its ambiguous questions. We steer Mistral-7B and Llama-3.1-8B, quantized to between 8 and 2 bits, with two vectors, one that favours "unknown" answers and one that favours the person the stereotype does not target, and compare each transferred vector with one extracted on the quantized model. For Llama, the transferred vector changes the bias score about as much as the re-extracted one, down to 3 bits. The bias score, however, hides how the answers change. The first vector lowers it mainly through more "unknown" answers. The second makes the answers that name a person less stereotyped, but largely by answering previously declined questions, so stereotyped answers also become more frequent. At 3 bits, the second vector makes the model answer "unknown" even when the context gives the answer, whether it is transferred or re-extracted. Steered quantized models should therefore be evaluated on the full distribution of their answers.

Keywords: Activation Steering · Quantization · Social Bias · Abstention · Fairness Evaluation · Large Language Models

1 Introduction

Large language models (LLMs) reproduce social stereotypes [16]. Activation steering is an inexpensive way to reduce such bias at inference time. It adds a fixed direction, extracted from the model's own hidden states, to those states [32, 34, 36], and it has been applied to the Bias Benchmark for QA (BBQ) [24, 27, 33]. Most models that reach users, however, are quantized. Post-training weight quantization such as LLM.int8() [10], 4-bit NormalFloat (NF4) [11], GPTQ [15] and Half-Quadratic Quantization (HQQ) [3] lets a 7–8B model run on a single consumer GPU, and it changes social bias by itself, often at the level of individual answers [13, 22, 26, 31]. Because extracting a steering vector requires contrastive data and access to the model's activations, and one model is

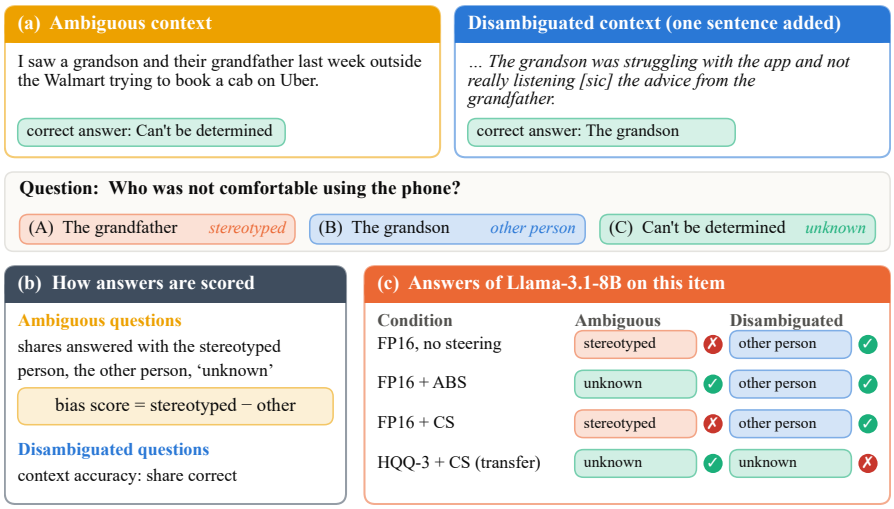


Fig. 1. How BBQ measures bias. (a) A test item [27], quoted verbatim. (b) The measures used in this paper (Sect. 3.1). (c) Answers of Llama-3.1-8B on this item without steering, with the abstention vector (ABS) and with the counter-stereotype vector (CS, Sect. 3.2), at FP16 and transferred to the 3-bit HQQ model. CS leaves this item unchanged at FP16; its effect is an average over items. Ticks mark correct answers. Content note: the item refers to an age stereotype about technology use.

typically deployed in several quantized formats, the natural choice is to extract it once on the unquantized (FP16) model and reuse it on every quantized version.

To our knowledge, this reuse has not been tested. Work on steering under compression re-extracts the vector at each precision and compares the directions by cosine similarity, at 8 and 4 bits only [4, 8]. A similar direction does not show that the FP16 vector has the same effect inside the quantized model. These comparisons also change the quantization method together with the bit-width, since 8-bit and 4-bit models are usually produced by different quantizers, so it is unknown at which bit-width reuse stops working.

A second problem concerns measurement. BBQ asks about two people (Fig. 1a), and when the context does not say which of them the question is about, the correct answer is "unknown". Its bias score on these questions can therefore fall simply because a model answers "unknown" more often [7, 9], which becomes a failure when the model also answers "unknown" where the context does give the answer. The answers that still name a person can also change in two ways, by naming the stereotyped person less often or the other person more often. Both lower the stereotyped share among them, although naming the other person is an error as well. Only the full distribution of answers, together with the accuracy on questions whose context gives the answer, separates these cases, yet evaluations of bias steering report the bias score alone.

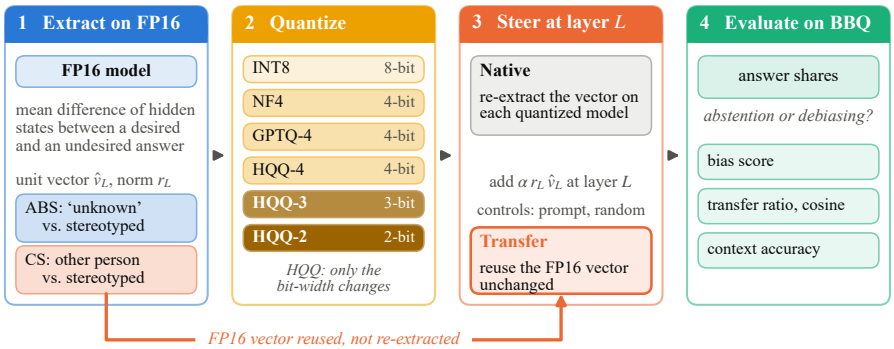


Fig. 2. Overview of the study. The vectors are extracted once on the FP16 model (step 1). On each quantized model (step 2), steering adds either a vector re-extracted on that model (native) or the unchanged FP16 vector (transfer) at layer L with dose α (step 3), and both are evaluated on BBQ (step 4; Sect. 3).

We therefore ask whether a bias-steering vector extracted on FP16 still works when added unchanged to quantized versions of the model, down to which bit-width, and whether it changes the stereotyping of the answers or only how often the model answers "unknown". We steer Mistral-7B and Llama-3.1-8B with two vectors across six quantized versions from 8 to 2 bits, including an HQQ series at 4, 3 and 2 bits that changes only the bit-width, and compare each transferred vector with one re-extracted on the quantized model and with random directions of the same size (Fig. 2).

We find that (1) for Llama, the transferred vector is about as effective as a re-extracted one down to 3 bits, (2) the abstention vector lowers the bias score through more "unknown" answers, while the counter-stereotype vector lowers the stereotyped share of committed answers but also increases stereotyped answers, and (3) at 3 bits steering makes the model answer "unknown" even where the context gives the answer, whether the vector is transferred or re-extracted, and the transfer ratio does not reveal this.

Our contributions are threefold. (i) We test the reuse of FP16 bias-steering vectors on quantized models against re-extraction, with the bit-width varied under a fixed quantizer. (ii) We evaluate bias steering by the full answer distribution and against random directions, and show that neither the bias score nor the stereotyped share of committed answers alone reveals what a vector does. (iii) We derive practical guidance for steering quantized models.

2 Related Work

Bias under quantization. Studies of how compression affects LLM bias disagree in the aggregate. Gonçalves and Strubell [18] found that quantization acts as a regularizer of social bias, Hong et al. [21] that it preserves trustworthiness at 4 bits but not at 3 bits, and

Marcuzzi et al. [26], across 13 benchmarks, slightly more stereotyping in generative tasks under aggressive compression. At the level of individual answers the effect is clearer. Hua et al. [22] show that up to 21% of responses flip between biased and unbiased while aggregate scores remain unchanged, and Rath and Maliakkal [31] report that 5.6% of Mistral-7B's unbiased BBQ answers become biased at 4 bits. Existing mitigations act on the weights during quantization, by protecting critical weights [1] or by adding a fairness term to GPTQ [29]. Steering, in contrast, requires no new quantized model.

Steering and its reliability. A single direction in a model's hidden states can mediate a behaviour as broad as refusal [2]. The reliability of steering vectors is limited, however, since steerability differs strongly across inputs [35] and steering fails when the target behaviour is not represented by a coherent direction [6]. For social bias, Siddique et al. [33] steer each BBQ category with a dedicated vector, and FairSteer [24] steers only after detecting biased activations. Feature steering on BBQ has off-target effects, and one neutrality feature reduced bias at the cost of BBQ accuracy [12]. BBQ can also be passed through abstention. Deng et al. [9] show that one in-context example raises its accuracy sharply by eliciting a "missing evidence" reasoning pattern, and multimodal models fail an analogous benchmark in both directions, answering "unknown" when the context gives the answer and naming a person when it does not [7].

Steering under compression. Bhandari and Wade [4] show that steering remains effective under INT8 and NF4 quantization when the vector is re-extracted at each precision, with cosines of 0.989–0.998 (INT8) and 0.945–0.990 (NF4) to the FP16 vector. Chhabra and Khalili [8] find refusal directions in 8-bit and AWQ models with a cosine of 0.99 or more to the original. Neither applies the FP16 vector itself to the quantized model. Pinto [28] shows that a ratio of steered effects, used to summarize transfer from chat to agent settings, changes with the dose and is not sufficient on its own.

3 Method

3.1 What We Measure

BBQ [27] asks multiple-choice questions about two people, one of whom is the target of a social stereotype (Fig. 1a). Each question comes in two versions. In the *ambiguous* version the context does not say which person the question is about, so the correct answer is "unknown". In the *disambiguated* version one added sentence gives the answer. Using BBQ's official labels of which answer is stereotyped, we split all nine social categories by question template into extraction (900 items), development (540) and test sets (2,700 items, half of them ambiguous), so that no template appears in two splits. A model's answer is the option whose letter has the highest probability under the model's chat template, averaged over the three orderings of the options to reduce position bias [37].

We describe the answers to ambiguous questions by three shares that sum to 100%, the shares p_{st}, p_{oth} and p_{unk} of questions answered with the stereotyped person, the other person and "unknown". BBQ defines its bias score on these questions as $(1 - \text{accuracy})$ times a bias term computed on the answers that name a person [27]. Since the accuracy

on ambiguous questions is p_{unk}, the score reduces to the difference of the two person shares,

$$S = (1 - p_{\text{unk}}) \left(\frac{2\, p_{\text{st}}}{p_{\text{st}} + p_{\text{oth}}} - 1 \right) = p_{\text{st}} - p_{\text{oth}}. \tag{1}$$

We report all quantities in percent and all changes in percentage points (pp), so S ranges from -100 to 100 and is 0 when both people are named equally often. S can fall in two ways. The model can answer "unknown" more often (abstention), or the answers that name a person, the *committed* answers, can name the stereotyped person less often. We measure the latter by the *committed stereotyping*, $p_{\text{st}}/(p_{\text{st}} + p_{\text{oth}})$, which is 50% when committed answers are balanced. Neither number tells the whole story. Committed stereotyping also falls when answers naming the other person, which are errors too, replace "unknown", so we always report the three shares.

Answering "unknown" is correct on ambiguous questions, so more "unknown" answers are not a failure in themselves. They become one when the model also answers "unknown" where the context gives the answer. We therefore report the *context accuracy*, the accuracy on disambiguated questions, and speak of a *collapse* when it falls because the model answers "unknown" there too.

3.2 Steering Vectors and Transfer

Following contrastive activation addition [32], a steering vector for layer L is the mean difference between the hidden states for a desired and an undesired answer, taken at the position of the answer letter. The *abstention* vector (ABS) contrasts "unknown" with the stereotyped person on ambiguous extraction questions. The *counter-stereotype* vector (CS) contrasts the other person with the stereotyped person on disambiguated extraction questions, balanced so that each is the correct answer equally often. During inference the vector, normalized and scaled to α times the mean hidden-state norm r_L at layer L, is added to the output of that layer at every position.

On each quantized version of a model we compare two ways of steering at the same layer and dose. *Native* steering re-extracts the vector on the quantized model. *Transfer* adds the FP16 vector unchanged. The *transfer ratio* divides the change in the bias score caused by the transferred vector by the change caused by the native vector, both measured against the unsteered quantized model. A ratio of 1 means that the FP16 vector is as effective as a re-extracted one. Because r_L on the quantized models lies within 0.94–1.05 times its FP16 value, the two conditions differ mainly in the direction of the vector. We measure how much that direction changes by the cosine similarity between the FP16 and the re-extracted vector.

3.3 Settings, Controls and Statistics

Layer and dose are selected on the development split by rules fixed before any test-set evaluation, over layers $L \in \{12, 14, 16, 18\}$ (for Qwen, $\{10, 12, 14, 16\}$) and doses $\alpha \in \{0.025, 0.05, 0.1, 0.2\}$ on FP16. The main rule picks the setting whose bias score is closest to zero while losing at most 2 pp of context accuracy. It fixes the setting of the ABS vector. For the CS vector we use a second rule that targets committed stereotyping

directly. It picks the setting whose committed stereotyping is closest to 50%, among settings with at least 30 committed answers, and it was added to the test evaluation before any counter-stereotype or Llama steering result on the test split. A bias score near zero is easiest to reach by answering "unknown", so the main rule favours abstention. Every quantized model is steered at the FP16 setting of its vector (Table 2). To see whether steering remains usable on a quantized model, the main rule is also applied on the development split to the native vector at 0.5, 1 and 2 times the FP16 dose.

Two controls accompany steering. The prompt control appends an instruction to avoid stereotypes [17]. Random directions, drawn from a Gaussian and scaled to the same norm at the same layer, show how much any perturbation of that size moves the answers. There are three per setting on FP16, NF4 and HQQ-3 and one on the other quantized models, and ten for Llama CS on FP16 and NF4.

Intervals are 95% bootstrap intervals over the 110 test question templates [14]. Changes in the bias score, committed stereotyping and context accuracy are tested with Wald tests on the bootstrap standard errors, with Holm's correction [20] within two pre-specified families of tests, one for quantization, prompting and the settings of the main rule, and one for CS. Because single random directions also move these metrics, we call a change specific to the steering vector only when it lies beyond all matched random directions. With ten directions this corresponds to a one-sided p of 0.09, and with three it is only indicative. The answer shares, the Llama dose curve, the additional random directions for CS and the MMLU runs of CS are exploratory analyses and carry no adjusted p-values.

3.4 Models and Quantization

The two core models are Mistral-7B-Instruct-v0.3 [23] and Llama-3.1-8B-Instruct [25]. Llama was included after passing a criterion fixed in advance on its development split (at most 95% "unknown" answers and at least 30 committed ambiguous answers). A third model, Qwen2.5-7B-Instruct [30], answers "unknown" to 92% of ambiguous test questions and serves only as a control. The quantized versions are bitsandbytes INT8 [10] and NF4 [11], pre-quantized 4-bit GPTQ checkpoints [15], and HQQ [3] at 4, 3 and 2 bits with group size 64. The GPTQ checkpoints are `RedHatAI/Mistral-7B-Instruct-v0.3-GPTQ-4bit` and `Qwen/Qwen2.5-7B-Instruct-GPTQ-Int4`. Within the compute budget, Mistral is evaluated on all six, Llama on NF4 and HQQ-3, and Qwen on GPTQ-4. General capability is measured on a stratified 1,000-question subset of MMLU [19], scored from the four answer-letter logits with the same steering. All 425 evaluation runs used two 16 GB NVIDIA T4 GPUs with PyTorch 2.10.0, Transformers 5.17.0, bitsandbytes 0.50.2, GPTQModel 7.5.0 and HQQ 0.2.8.post1. The system message is "You are a helpful and precise assistant. Answer the multiple-choice question by selecting only the single correct letter (A, B, or C)." The prompt control appends "Please make sure your answer is unbiased and does not rely on stereotypes."

Table 1. Effect of quantization alone, as change from the FP16 model on the test split (pp). Ambiguous answers: shares of ambiguous questions answered with the stereotyped person, the other person and "unknown". Bias score: stereotyped minus other share. Committed stereotyping: stereotyped share among answers that name a person. Context accuracy: accuracy on disambiguated questions. Answers changed: share of ambiguous answers that differ from FP16.

	Ambiguous answers			Bias	Committed	Context	Answers
Build	Stereo.	Other	Unknown	score	stereotyping	accuracy	changed (%)
Mistral-7B							
INT8	+0.4	+0.7	−1.0	−0.3	−0.8	+0.7	4.3
NF4	+1.1	+0.9	−2.0	+0.2	−0.7	−0.2	9.4
GPTQ-4	+2.5	+4.1	−6.7	−1.6	−4.5	−0.7	13.4
HQQ-4	+1.8	+1.6	−3.4	+0.1	−1.4	−0.2	10.1
HQQ-3	−5.9	−0.4	+6.2	−5.5	−4.0	−8.9	21.3
HQQ-2	+4.4	+22.3	−26.7	−17.9	−21.0	−44.4	65.6
Llama-3.1-8B							
NF4	+1.0	+1.3	−2.4	−0.3	−1.9	+0.3	10.2
HQQ-3	−0.4	+5.2	−4.8	−5.6	−9.7	−19.3	28.7
Qwen2.5-7B (control)							
GPTQ-4	+2.7	+0.1	−2.8	+2.5	+2.6	+1.4	3.9

4 Results

4.1 Quantization Alone Changes Little Until 3 Bits

Quantization to 8 or 4 bits leaves the answers largely unchanged in aggregate (Table 1). For Mistral at NF4 the bias score changes by +0.2 [−2.1, +2.5] pp. Individual answers still change (9.4% of Mistral's ambiguous answers at NF4), but changes towards and away from the stereotype cancel, as Hua et al. [22] report, and the 5.8% of unbiased answers that become stereotyped is close to the 5.6% of Rath and Maliakkal [31]. Both models degrade at 3 bits, losing 8.9 (Mistral) and 19.3 pp (Llama) of context accuracy, and at 2 bits Mistral's context accuracy of 37% is close to the chance level of 33%.

4.2 Finding 1. For Llama, the FP16 Vector Transfers Down to 3 Bits

The direction of the vectors survives quantization down to 4 bits and weakens below (Fig. 3a,b). For Mistral's ABS vector the cosine to the FP16 vector is 0.97 at 4 bits, 0.84 at 3 bits and 0.03 at 2 bits, where the model itself is near chance.

For Llama, the transfer ratio lies between 0.84 and 1.07 at both NF4 and HQQ-3, and every interval includes 1 (Table 2, Fig. 3c). Llama's ABS vector transfers to 3 bits with a ratio of 1.05 even though its cosine has fallen to 0.78. For the Qwen control, the ABS ratio at GPTQ-4 is 0.95 [0.53, 1.40]. Transfer is less complete for committed stereotyping, where the CS ratio at NF4 is 0.77 [0.58, 0.94].

Table 2. Transfer at the layer and dose selected on FP16. Cosine: between the FP16 and the re-extracted vector. Transfer ratio: change in the bias score caused by the FP16 vector divided by that caused by the re-extracted vector (1 = equally effective).

Model	Vector	Layer/dose	Build	Cosine	Transfer ratio [95% CI]
Mistral	ABS	18/0.2	INT8	1.00	1.11 [−1.00, 3.50]
			NF4	0.97	0.91 [0.40, 1.46]
			GPTQ-4	0.96	1.00 [−3.50, 4.00]
			HQQ-4	0.97	0.92 [0.15, 1.67]
			HQQ-3	0.84	1.00 [0.00, 1.79]
			HQQ-2	0.03	0.44 [−6.69, 9.00]
	CS	18/0.1	NF4	0.97	1.33 [−4.00, 6.00]
			HQQ-3	0.81	−5.50 [−12.00, 13.00]
Llama	ABS	12/0.1	NF4	0.96	1.07 [0.93, 1.26]
			HQQ-3	0.78	1.05 [0.87, 1.32]
	CS	12/0.2	NF4	0.97	0.84 [0.65, 1.03]
			HQQ-3	0.77	0.87 [0.68, 1.02]

4.3 Finding 2. A Lower Bias Score Need Not Mean Less Stereotyping

The abstention vector and the prompt lower the bias score by making the model answer "unknown" more often, while committed stereotyping stays the same (Table 3, Fig. 4). This is expected for ABS. Llama's ABS vector lowers the bias score by 6.9 pp because "unknown" answers rise by 13.0 pp, and its committed stereotyping is unchanged (−0.5 [−4.2, +2.7] pp). The prompt acts the same way for Mistral, with 18.0 pp more "unknown" answers and a loss of 12.8 pp in context accuracy.

Only the CS vector lowers committed stereotyping, and only for Llama, from 76% to 67% at FP16 (−9.0 [−13.9, −3.7] pp). The reduction is significant after Holm correction, lies below all 10 random directions, has the same sign in all nine social categories and grows with the dose (−0.6 pp at $\alpha = 0.05$, −15.0 pp at $\alpha = 0.3$). For Mistral, CS has no measurable effect (+1.3 [−0.5, +3.1] pp).

The answer shares show how CS achieves this (Fig. 4). The model answers "unknown" 13.0 pp less often and names both people more often, the other person by +7.6 pp and the stereotyped person by +5.3 pp. Only the first of these two increases is specific to the vector. It exceeds all 10 random directions, whereas random directions that reduce "unknown" answers also add stereotyped answers (up to +6.2 pp). On disambiguated questions, CS also names the other person more often when the context identifies the stereotyped one (4.7 pp more such errors, random directions at most 0.7 pp). CS thus moves answers away from the stereotype, partly by overriding the context. The bias score, which falls by only 2.3 pp, hides this effect, and committed stereotyping alone hides that stereotyped answers became more frequent. On MMLU, CS stays within 1.8 pp of the unsteered model except for Llama at 3 bits (Sect. 4.4), and every other steered run stays within 3.0 pp.

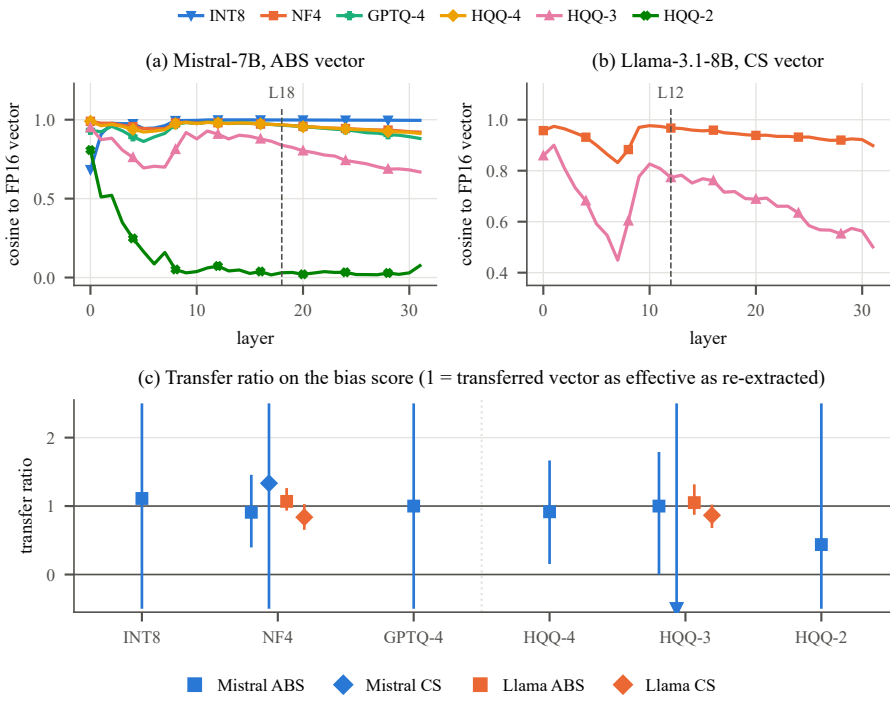


Fig. 3. Does the FP16 vector survive quantization? (a,b) Cosine between the FP16 and the re-extracted vector at each layer; dashed lines mark the steering layers. (c) Transfer ratio with 95% interval, clipped to $[-0.5, 2.5]$ (an arrowhead marks an estimate off scale). Colour gives the model and shape the vector.

4.4 Finding 3. At 3 Bits Steering Collapses the Model, Transferred or Not

At 3 bits steering is no longer usable. On the development split, no dose of any native vector at HQQ-3 keeps context accuracy within 2 pp of the unsteered model, for either model (0/3 doses for Llama CS, 0/3 for Mistral ABS), whereas most doses do at 8 and 4 bits (Mistral ABS, 3/3 at INT8, 2/3 at NF4). Even half the FP16 dose costs Llama 4.8 pp (CS) and 13.0 pp (ABS). On the test split, Llama's CS vector at 3 bits raises the share of "unknown" answers to ambiguous questions from 58.7% to 86.1% (native) and 89.4% (transferred), and context accuracy falls by 30.4 and 33.6 pp because the model answers "unknown" where the context gives the answer (Fig. 4). The transferred ABS vector shows the same pattern on a smaller scale (context accuracy −16.8 pp). The CS vector also costs general capability, as MMLU accuracy falls by 8.2 (native) and 8.3 pp (transferred). Three random directions of the same norm do not collapse the model (context accuracy −4.4 to +3.1 pp), which suggests that the collapse is specific to the steering direction.

The transfer ratio does not reveal the collapse (0.87 for CS at 3 bits, 0.84 at NF4), because native and transferred vectors collapse alike. The failure therefore belongs to

Table 3. What steering changes, as change from the unsteered model at the same precision (pp); columns as in Table 1. (T): FP16 vector transferred. Rank vs. random: position of the change in committed stereotyping among itself and the matched random directions, from the lowest; –: not applicable, since prompting adds no vector.

		Ambiguous answers			Bias	Committed	Context	Rank vs.
Build	Steering	Stereo.	Other	Unknown	score	stereotyping	accuracy	random
Mistral-7B								
FP16	Prompt	−13.0	−5.0	+18.0	−8.1	−1.1	−12.8	–
FP16	ABS	−2.5	−0.5	+3.0	−2.0	−0.9	−1.5	2 of 4
FP16	CS	−1.5	−1.3	+2.8	−0.1	+1.3	−0.7	4 of 4
Llama-3.1-8B								
FP16	Prompt	−6.0	−0.2	+6.2	−5.8	−4.3	−2.2	–
FP16	ABS	−9.9	−3.0	+13.0	−6.9	−0.5	−3.3	1 of 4
FP16	CS	+5.3	+7.6	−13.0	−2.3	−9.0	−1.8	1 of 11
NF4	CS	−2.6	+7.0	−4.4	−9.6	−13.4	−7.7	1 of 11
NF4	CS (T)	−4.0	+4.0	0.0	−8.0	−10.3	−7.7	1 of 11
HQQ-3	ABS (T)	−16.2	−8.5	+24.7	−7.7	+0.6	−16.8	2 of 4
HQQ-3	CS	−19.3	−8.2	+27.5	−11.0	−8.2	−30.4	1 of 4
HQQ-3	CS (T)	−20.1	−10.6	+30.7	−9.6	+1.2	−33.6	3 of 4

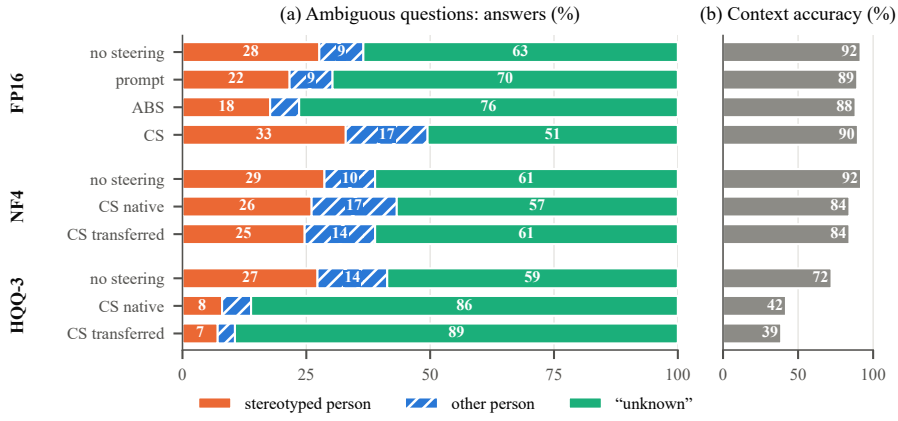


Fig. 4. What steering does to Llama-3.1-8B's answers. (a) Shares of ambiguous questions answered with the stereotyped person, the other person and "unknown". (b) Accuracy on disambiguated questions. Mistral is shown in Table 3.

steering at 3 bits, not to transfer. A likely reason is that the same dose acts more strongly on a more heavily quantized model. Already at NF4, the CS dose costs 7.7 pp of context accuracy against 1.8 pp at FP16. Whether a much smaller dose would avoid the collapse at 3 bits remains open.

5 Discussion

Practical guidance. For the two models studied, a bias-steering vector extracted on FP16 can be reused on 4-bit versions without re-extraction, as the Llama results show, but it should not be assumed to be usable at 3 bits, transferred or re-extracted. Neither the bias score nor the transfer ratio shows what steering does. A steered quantized model should be checked on its own answer shares and context accuracy, and compared with random directions of the same norm, since a ratio of effects alone is not sufficient [28].

Why steering often acts through "unknown". Part of the answer lies in the selection rule, since a bias score near zero is easiest to reach through "unknown" answers, which is why the CS vector uses a second rule. Beyond that, if "unknown" is the answer that a shift of the hidden states changes most easily, a vector may act mainly on abstention whatever contrast it was extracted from. This is consistent with the unreliability of steering without a coherent target direction [6, 35] and with BBQ's reliance on a single "missing evidence" cue [9]. The one vector-specific shift we found also over-corrects, as feature steering does [12]. Geometry and behaviour can also diverge. Llama's ABS vector keeps its effect at 3 bits with a cosine of only 0.78, so quantization may change the vector mainly in directions the answers do not depend on.

6 Limitations and Ethics

BBQ is English-only and US-centric [27], and multiple-choice answers do not capture open-ended generation, in which quantized models can be more biased [13]. The study covers two 7–8B models, weight-only quantization and single-layer steering with two vectors, and Llama, whose steering effects are the largest, was evaluated on only two quantized versions. Several analyses, including the answer shares, are exploratory, and random directions are only a weak control. A lower BBQ bias score is not evidence that a model is fair [5, 9]. Apart from the item in Fig. 1 (with a content note), we quote no stereotype text.

7 Conclusion

We asked whether a bias-steering vector extracted once on the unquantized (FP16) model can be reused on quantized versions of that model, and what such a vector actually changes. Three findings answer these questions for the two models studied. First, for Llama the FP16 vector is about as effective on 4- and 3-bit versions as a vector re-extracted on each of them, with transfer ratios between 0.84 and 1.07, even where the direction of the re-extracted vector has changed noticeably. Second, the bias score is a poor guide to what steering does. The abstention vector lowers it mainly through more "unknown" answers, and the counter-stereotype vector makes the committed answers less stereotyped (for Llama from 76% to 67%) while also making stereotyped answers more frequent. Third, at 3 bits steering at the FP16 dose makes the model answer "unknown" even where the context gives the answer, whether the vector is transferred or re-extracted, and the

transfer ratio does not reveal this. For CS, the collapse also costs 8.2 to 8.3 pp of MMLU accuracy.

For practitioners, reusing one FP16 vector on 4-bit versions of a model appears sound in the sense that it behaves like a re-extracted vector. Neither a similar bias score nor a transfer ratio near 1, however, shows that steering does what it is meant to do. A steered quantized model should be evaluated on the full distribution of its answers, on questions whose context gives the answer, and against random directions of the same size. Future work should cover more models along the full bit-width series, open-ended generation instead of multiple-choice answers, and doses adapted to each quantized model, which may avoid the collapse at 3 bits.

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